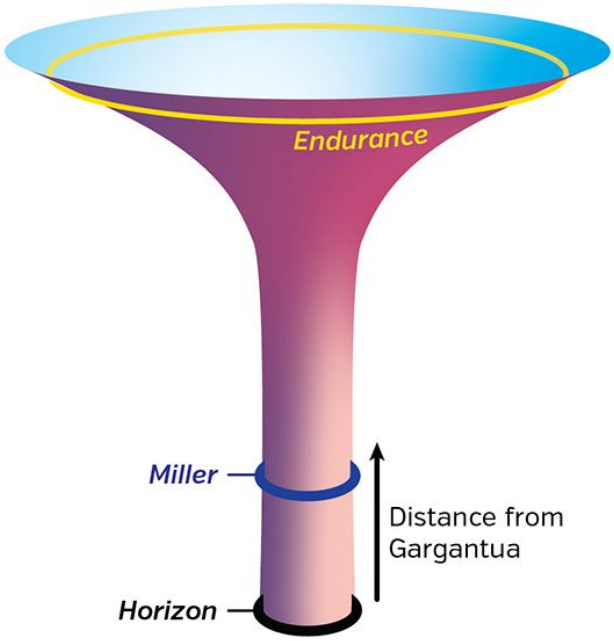


Fig. 17.1. The warped space around Gargantua as seen from the bulk, with one space dimension omitted. Also, the orbits of Miller's planet and the *Endurance*, parked and waiting for the crew to return.

Space there is warped like the surface of a cylinder. In the figure, the cylinder's cross sections are circles whose circumferences don't change as we move nearer to or farther from Gargantua. In reality, when we restore the missing dimension, the cross sections are spheroids, whose circumferences don't change as we move nearer or farther.

So why is this location different from any other on the cylinder? What makes this location special?

The key to the answer is the warping of time, which does *not* show up in Figure 17.1. Time slows near Gargantua, and the slowing becomes more extreme as we get closer and closer to Gargantua's event horizon. Therefore, according to Einstein's law of time warps (Chapter 4), gravity becomes ultrastrong as we near the horizon. The red curve in Figure 17.2, which depicts the strength of the gravitational force, turns sharply upward. By contrast, the centrifugal force that the planet feels